Reconciliation of DFD Mass Models A and B: Exponents, Prefactors, and the Bottom Quark

DFD Cross-Reference Analysis

23 March 2026

1 Statement of the Problem

Two mass models exist in the DFD codebase, both implementing the master formula

$$m_f \;=\; A_f\,\alpha^{n_f}\,\frac{v}{\sqrt{2}}\,, \qquad \alpha = \tfrac{1}{137.036}, \quad \frac{v}{\sqrt{2}} = 174.1\ \text{GeV}. \tag{1}$$

Model A (v67 dictionary, January 2026 — Gary's verified ground truth; file `dfd_v67_mass_dictionary_verify.py` and `compute_all_masses.py`). Mean error 1.42%.

Model B (March 2026 agent-derived; file `compute_mass_exponents.py`). Mean error 1.73%.

They agree on 7 of 9 fermions but diverge on **two down-type quarks**:

	Exponent n_f		Prefactor A_f		% Error	
Fermion	A	B	A	B	A	B
b	0	1	1/42	π	−0.83	−4.51
d	2.5	2.0	6	1/2	+1.75	−0.74

All other fermions $(e, \mu, \tau, u, c, t, s)$ share identical exponents, though some prefactors differ.

2 Question 1: Can the Formula $n_f = (k_f + k_H)/2$ Reproduce Both?

Model B derives all 9 exponents from

$$n_f = \frac{k_f + k_H}{2}\,, \qquad k_H = \begin{cases} +1 & \text{leptons, down quarks (H-coupling)} \\ -1 & \text{up quarks } (\tilde{H}\text{-coupling}) \end{cases} \tag{2}$$

with three type-dependent rules for k_f:

$$\text{Leptons:} \quad k_f = 2^{3-g}\,, \tag{3}$$

$$\text{Down quarks:} \quad k_f = 4 - g\,, \tag{4}$$

$$\text{Up quarks:} \quad k_f = T(4-g) = \tfrac{(4-g)(5-g)}{2}\,. \tag{5}$$

Reverse-engineering Model A. We solve $k_f = 2n_f - k_H$ for every fermion:

Fermion	k_H	$n_f^{(A)}$	$k_f^{(A)}$	$n_f^{(B)}$	$k_f^{(B)}$
e	$+1$	2.5	4	2.5	4
μ	$+1$	1.5	2	1.5	2
τ	$+1$	1.0	1	1.0	1
u	-1	2.5	6	2.5	6
c	-1	1.0	3	1.0	3
t	-1	0.0	1	0.0	1
d	$+1$	2.5	4	2.0	3
s	$+1$	1.5	2	1.5	2
b	$+1$	0.0	-1	1.0	1

Result. Model A requires $k_f(b) = -1$ and $k_f(d) = 4$. The first is **geometrically impossible**: holomorphic line bundles $\mathcal{O}(k)$ on $\mathbb{CP}^2$ require $k \geq 0$, and the Dirac index formula $\mathrm{ind}(D \otimes \mathcal{O}(k)) = (k+1)(k+2)/2$ has no zero modes for $k < 0$. The value $k_f(d) = 4$ is permissible but clashes with the linear rule $k_f = 4 - g$ (which gives 3, not 4, for $g = 1$).

Answer: No. Model A's down-quark exponents *cannot* be reproduced by (**??**) with any valid k_f assignments. Model A therefore does *not* derive its exponents from the spin^c bundle formula; it treats the exponent set $\{2.5, 1.5, 0\}$ as phenomenological, with the $n_b = 0$ value attributed to "QCD dressing" (see line 92 of `compute_all_masses.py`).

3 Question 2: Physical Meaning of $n_b = 0$ (Model A)

In Model A the bottom quark (b, 3rd generation, down-type) has $n_b = 0$, meaning

$$m_b = A_b \cdot \alpha^0 \cdot \frac{v}{\sqrt{2}} = \frac{1}{42} \cdot 174.1 \text{ GeV} = 4.145 \text{ GeV}.$$

The bottom quark mass is set by the **electroweak scale** v **alone**, suppressed only by the prefactor $A_b = 1/42$.

Physics of $n_b = 0$ and $n_t = 0$ together. Both 3rd-generation quarks (t and b) sit at the "zero-suppression" level. This means neither receives any α-power suppression from gauge dressing. Physically:

- The top quark ($A_t = 1$, $n_t = 0$) has mass $\approx v/\sqrt{2}$: it is unsuppressed in *both* prefactor and exponent.
- The bottom quark ($A_b = 1/42$, $n_b = 0$) is unsuppressed in exponent but has a huge QCD prefactor suppression.
- The ratio $m_t/m_b = A_t/A_b = 42$ is thus explained *entirely* by the QCD running factor $N_f \cdot b_0 = 6 \times 7 = 42$.

The interpretation: in Model A, the t–b mass splitting is a pure QCD effect, not a geometric (bundle-degree) effect. Both live at the base level of the Cayley-graph propagation hierarchy.

4 Question 3: Physical Meaning of $n_b = 1$ (Model B)

In Model B the bottom quark has $n_b = 1$, so

$$m_b = \pi \cdot \alpha^1 \cdot \frac{v}{\sqrt{2}} = \pi \cdot 1270.5 \text{ MeV} = 3991 \text{ MeV}.$$

This is a 4.5% error from the observed 4180 MeV.

Physics of $n_b = 1$ versus $n_t = 0$. Here b and t are *not* at the same level despite being the same generation. The b quark receives one power of α suppression that the t quark does not. Physically this means:

- The t–b splitting comes from *geometry* ($n_t = 0$ vs $n_b = 1$), not from QCD running.
- The down-quark sequence $\{n_d, n_s, n_b\} = \{2, 1.5, 1\}$ follows the linear rule $k_f = 4-g$ identically to the lepton-like pattern shifted by one unit.
- All 3rd-generation fermions with $k_H = +1$ (τ, b) share $k_f = 1$ and hence $n_f = 1$, while the 3rd-generation fermion with $k_H = -1$ (t) gets $n_f = 0$.

The interpretation: in Model B, the t–b splitting is a **Higgs-coupling geometric effect**. The top couples to $\tilde{H}$ ($k_H = -1$) while the bottom couples to H ($k_H = +1$), and this single sign flip creates the one-unit exponent difference.

5 Question 4: Which Model Is Better Supported by A_5 Geometry?

5.1 Arguments favouring Model B (geometric consistency)

1. **All $k_f \geq 0$.** Model B assigns $k_f \in \{1, 2, 3\}$ for down quarks, all of which are valid line-bundle degrees on $\mathbb{CP}^2$. Model A requires $k_f = -1$ for b, which is geometrically inadmissible.
2. **Uniform derivation.** Model B derives all 9 exponents from three simple rules (doubling, linear, triangular), each with a clear A_5 Cayley-graph origin. Model A must invoke an ad hoc "QCD dressing" mechanism to shift n_b from 1 to 0.
3. **Index theorem consistency.** At $k_f = 1$, the Atiyah–Singer index gives ind $= 3$, which places t, τ, and b together at the base level—exactly three fermions per $k_f = 1$, matching $N_{\text{gen}} = 3$.
4. **Structural elegance.** The down-quark d quark prediction is actually *better* in Model B (-0.74% vs $+1.75\%$).

5.2 Arguments favouring Model A (phenomenological accuracy)

1. **Superior b quark accuracy.** Model A: -0.83% error; Model B: -4.51% error. This is a 3.7 percentage-point advantage, and the b quark has a precisely measured mass.
2. **Lower mean error.** Model A achieves 1.42% mean error versus 1.73% for Model B.
3. **Algebraically closed prefactors.** Model A derives every prefactor from the operator chain $A_f = G(g) \cdot S_f(g)$ with $G = \text{diag}(2/3,\ 1,\ 1)$, $Q_d = \text{diag}(1,\ 6/7,\ 1/42)$, and $K_d = J_3$, $K_u = I_4$. Every constant traces to either the A_5 Cayley structure or QCD β-function coefficients. Model B's prefactors ($2/\pi$, $2\sqrt{2}$, π, $\sqrt{3}/2$, $1/2$) lack a comparably detailed derivation chain.

4. **Structural ratio verification.** Model A verifies six non-trivial ratios ($A_d/A_u = 9/4$, $A_t/A_b = 42$, etc.) that all pass exactly. These ratios encode J_3 vs I_4 kernel structures and QCD running.

5.3 Verdict

Model B has the more rigorous geometric derivation of exponents. It is the only model where all 9 exponents follow from the spin^c bundle formula with $k_f \geq 0$.

Model A has the more rigorous algebraic derivation of prefactors and achieves better phenomenological accuracy, particularly on the b quark.

The tension is real: Model A's exponents are phenomenologically superior but geometrically inconsistent (requiring $k_f = -1$), while Model B's exponents are geometrically rigorous but phenomenologically inferior on the b quark.

6 Question 5: Can the Two Models Be Unified?

6.1 The core conflict

The models are *not* trivially reconcilable. They disagree on a binary structural question: does the b quark sit at exponent level 0 or 1? There is no continuous parameter that interpolates between these.

6.2 A proposed unified framework

The natural resolution is to **adopt Model B's exponents with Model A's prefactor algebra**, modified as follows:

Step 1: Accept the geometric exponents. Use Model B's spin^c-derived exponents for all 9 fermions:

$$n_f = \frac{k_f + k_H}{2}\,, \qquad k_f^{\text{down}} = 4 - g\,,$$

giving $n_b = 1$, $n_d = 2$.

Step 2: Re-derive prefactors. The master formula becomes

$$m_b = A_b^{(\text{unified})} \cdot \alpha \cdot \frac{v}{\sqrt{2}}\,.$$

Requiring $m_b = 4180$ MeV gives

$$A_b^{(\text{unified})} = \frac{4180}{1270.5} = 3.290\,.$$

The question is whether this admits a closed-form expression from the $A_5 \times$ QCD operator algebra.

Candidate: $A_b = 42\alpha$. Since $A_b^{(A)} = 1/42$ with $n_b^{(A)} = 0$, the "unified" prefactor absorbs the exponent shift:

$$A_b^{(A)} \cdot \alpha^0 = A_b^{(\text{uni})} \cdot \alpha^1 \quad \Longrightarrow \quad A_b^{(\text{uni})} = \frac{A_b^{(A)}}{\alpha} = \frac{1}{42\alpha} = \frac{137.036}{42} = 3.263\,.$$

This gives $m_b = 3.263 \times 1270.5 = 4145$ MeV (same as Model A, -0.83% error), but the prefactor $1/(42\alpha)$ is dimensionally correct yet structurally problematic: it re-introduces α into the prefactor, defeating the purpose of having α only in the exponent.

Candidate: $A_b = G(3) \cdot R_d(3) \cdot \beta_{\mathbf{QCD}}$**.** A more principled approach: keep $G = \mathrm{diag}(2/3,\, 1,\, 1)$ and $K_d = J_3$ from Model A but replace the QCD running operator $Q_d = \mathrm{diag}(1,\, 6/7,\, 1/42)$ with a version adapted to the $n_b = 1$ exponent level. At $n_b = 1$, the b quark mass runs from scale $\alpha v/\sqrt{2} \approx 1.27$ GeV rather than $v/\sqrt{2} \approx 174$ GeV. The QCD running factor between 1.27 GeV and $m_b = 4.18$ GeV is substantially different from the running factor between 174 GeV and m_b. A careful re-evaluation of $Q_d(3,3)$ at this lower scale could yield a prefactor near 3.29.

Candidate: $A_b = (2/3) \cdot (7/\sqrt{3}) \cdot R_d(3)$**.** With $R_d(3) = 1$, this gives $A_b = 14/(3\sqrt{3}) = 2.694$, which is too small. No simple rational combination of A_5 group-theoretic numbers reaches 3.29 exactly.

6.3 Assessment

A clean unification requires **re-deriving the QCD running operator Q_d at the appropriate energy scale set by the exponent**. If $n_b = 1$ (Model B), then $Q_d(3,3)$ should encode running from $\mu \sim \alpha\, v/\sqrt{2}$ rather than $\mu \sim v/\sqrt{2}$. This is a calculable quantity in perturbative QCD but has not been carried out in either model.

The unified framework would then be:

1. Exponents from Model B (geometric, all $k_f \geq 0$);
2. Generation operator G from Model A (Cayley-graph origin);
3. Kernel operators $K_d = J_3$, $K_u = I_4$ from Model A;
4. QCD running operator Q_d **re-derived** at the scale set by $\alpha^{n_f}\, v/\sqrt{2}$ for each fermion.

7 Summary of Findings

Criterion	**Model A**	**Model B**
Mean error	1.42%	1.73%
b quark error	−0.83%	−4.51%
d quark error	+1.75%	−0.74%
Geometric consistency ($k_f \geq 0$)	No ($k_f = -1$)	Yes
Exponents from first principles	No (QCD dressing ad hoc)	Yes
Prefactors from first principles	Yes	No (asserted)
Structural ratio verification	6/6 pass	Not tested

Bottom line.

- Model A's $n_b = 0$ is **phenomenologically better** but **geometrically inadmissible** under the spin^c bundle formula.

- Model B's $n_b = 1$ is **geometrically required** but produces a 4.5% error on b with the current prefactor $A_b = \pi$.
- The path to unification is clear: **adopt Model B's exponents and re-derive the prefactors** (especially Q_d) at the correct energy scale. This should recover Model A's accuracy while maintaining geometric rigour.
- The b quark is the single decisive test case. Getting $A_b \approx 3.29$ from the $A_5 \times$ QCD operator algebra at scale $\mu \sim 1.27$ GeV would close the gap.